XINGYUE FENG

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Personal Profile

Ph.D. Student, Major in **Computer Science (CS)**

Research Interests: Computational Biology & Bioinformatics, AI for Structural Biology, Protein Structure Prediction, Protein Dynamics and Trajectory Analysis, Protein Function Prediction.

Personal Website: https://xingyue-feng.github.io/

EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) Blacksburg, VA
Ph.D. in Computer Science Sept 2025-Now

- Advisor: Dr. Debswapna Bhattacharya
- Reserch Area: Computational biology, AI for Molecular modeling

Northeast Normal University (NENU) Changchun, China
B.S. Computer Science and Technology Sept 2021-Jun 2025

- GPA: 3.85/4.0
- First-Class Scholarship, <5% (2022 &2024 &2025)

RESEARCH EXPERIENCE

Bhattacharya Lab, Virginia Tech Blacksburg, VA
Ph.D. Student | Mentor: Prof. Debswapna Bhattacharya Sept 2025-Now

Project 1. Protein Structure Prediction

- Developed Alpha3DiFold, a sequence-to-3Di prediction framework that enhances AlphaFold3 through structure-aware template retrieval, improving protein structure prediction for monomers and multimers. Poster presented at ISMB 2026.

Project 2. Protein Binding Site Prediction

- Co-developed xBind, an integrated web server for protein – protein, protein – DNA, and protein – RNA binding site prediction, supporting automated inference and interactive structural visualization. Co-first-author publication in Nucleic Acids Research (2026).

Project 3. Protein Function Prediction and Folding Dynamics

- Developing computational methods to predict protein GO terms and investigate the formation of structural interactions during protein folding.

Kihara Lab, Purdue University West Lafayette, IN
Undergraduate Researcher | Mentor: Prof. Daisuke Kihara Jul 2024-Feb 2025

Project 1. Protein Ligand Docking

- Building a Schrödinger-based docking pipeline to study protein–ligand interactions and evaluate how template-derived irons affect binding outcomes.

Project 2. Molecule (protein, DNA) Analysis & Design

- Modified Chroma to design cyclic peptides with specific ring structures, introducing a sequence renumbering mechanism for improved modeling flexibility.
- Developed deep generative models to design DNA sequences that bind target proteins, showing notable

improvements in sequence quality and hotspot targeting.

Upscaling Mouse Spinal Cord Atlas for 3D Alignment Remote
Participant | Mentor: Prof. Guangwei Zhang, University of Southern California Jun 2023

- Developed methods to upscale and align mouse spinal cord atlases from 2D slices into refined 3D models, resulting in a publication at ICBBS 2024.

Crowd Accident Monitoring and Early Warning System Based on Digital Twins and CNN Changchun, China
Participant | Mentor: Prof. Huijie Zhang, NENU Dec 2022-Dec 2023

- Developed a system for crowd analysis by combining YOLO-based detection, DeepSORT tracking, and HGNN modeling with 3D scene reconstruction and visualization, supported by a grant from the Jilin Students' Innovation and Entrepreneurship Training Program.

SKILLS

Programming: Python(4 years), Java(3 years), C, C#
Tools: SpringBoot, MySQL, Nginx, Unity, OpenGL, Blender, Autodock, Schrodinger Suite

ACTIVITIES

Bluemsun Studio, NENU | *Backend Developer* Aug 2022-Mar 2024

- Maintained various websites, including the University's online supermarket, the grantee management system, and the China Women's University student archive management system
- Supported outsourced administrative projects

"Create Your Future" Youth Programming Training Program | *Volunteer Tutor* Jul 2022-Aug 2022

- Assisted ethnic minority primary students in Xinjiang with learning programming, solving programming-related problems, and completing online courses